

Translated

From: e-maxx.ru

Maximum flow - Dinic's algorithm

Dinic's algorithm solves the maximum flow problem in $O(V^2E)$. The maximum flow problem is defined in this article [Maximum flow - Ford-Fulkerson and Edmonds-Karp](#). This algorithm was discovered by Yefim Dinitz in 1970.

Definitions

A **residual network** G^R of network G is a network which contains two edges for each edge $(v, u) \in G$:

- (v, u) with capacity $c_{vu}^R = c_{vu} - f_{vu}$
- (u, v) with capacity $c_{uv}^R = f_{vu}$

A **blocking flow** of some network is such a flow that every path from s to t contains at least one edge which is saturated by this flow. Note that a blocking flow is not necessarily maximal.

A **layered network** of a network G is a network built in the following way. Firstly, for each vertex v we calculate $level[v]$ - the shortest path (unweighted) from s to this vertex using only edges with positive capacity. Then we keep only those edges (v, u) for which $level[v] + 1 = level[u]$. Obviously, this network is acyclic.

Algorithm

The algorithm consists of several phases. On each phase we construct the layered network of the residual network of G . Then we find an arbitrary blocking flow in the layered network and add it to the current flow.

Proof of correctness

Let's show that if the algorithm terminates, it finds the maximum flow.